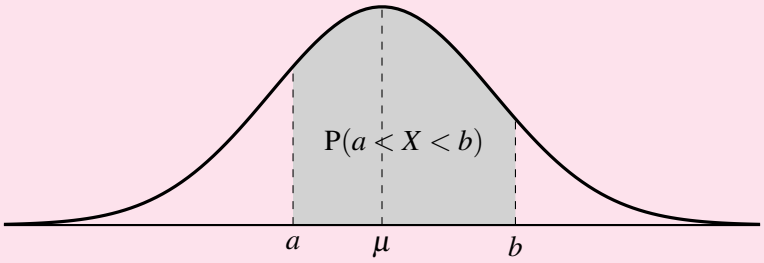
Normal Distribution

IB Maths AA SL/HL 4.9 · Notes · Questions · Worked Answers

1. Properties, symmetry & the 68–95–99.7 rule
2. Finding probabilities
3. Inverse normal & unknown μ or σ
4. Normal with binomial, conditional & expected numbers



No calculator Calculator (TI-Nspire CX II) allowed

Every item in the Index is a clickable link. Each topic has its own colour.

Index

= no calculator = calculator allowed

Sources: **Worksheet** = original *Normal Distribution* worksheet questions; **RV** = Revision Village practice questions (Practice Set 2 numbering); **SME** = Save My Exams IB *Normal Distribution*; **Oxford** = Oxford IB textbook Ex 14J and exam-style questions; **Y13 Worksheet** = IB Year 13 Binomial & Normal worksheet; **Kognity** = Kognity practice sets. Questions on finding μ, σ or a boundary value are mainly in the *Inverse Normal* booklet. Repeated questions appear only once.

Key Notes (click any line to jump; every page has a link back here)

1. What is a normal distribution? (continuous) . . . p. 4
2. The 68–95–99.7 rule . . . . . p. 4
3. Standardising: the z-score . . . . . p. 4
4. Using the TI-Nspire CX . . . . . p. 5
5. Always sketch! . . . . . p. 5
6. Using symmetry (no calculator) . . . . . p. 5
7. Inverse normal: use the left tail . . . . . p. 5
8. Unknown μ or σ . . . . . p. 6
9. Normal + binomial; expected numbers . . . . . p. 6
10. Conditional probability . . . . . p. 6

Sub-topics

Section 1: Properties, Symmetry & the 68–95–99.7 Rule . . . . . p. 7 (8 questions)
Section 2: Finding Probabilities . . . . . p. 15 (29 questions)
Section 3: Inverse Normal & Unknown μ or σ . . . . . p. 41 (9 questions)
Section 4: Normal with Binomial, Conditional & Expected Numbers . . . . . p. 51 (14 questions)

Questions

1. Properties, Symmetry & the 68–95–99.7 Rule

1. RV · Q2 . . . . . Q 7 · A 8
2. RV · Q3 . . . . . Q 7 · A 8
3. Worksheet · Q1 . . . . . Q 9 · A 10
4. Worksheet · Q2 . . . . . Q 9 · A 10
5. Worksheet · Q8 . . . . . Q 11 · A 12
6. Kognity B · Q3 . . . . . Q 11 · A 12
7. Kognity B · Q4 . . . . . Q 11 · A 12
8. SME Medium · Q1 . . . . . Q 13 · A 14

2. Finding Probabilities

9. RV · Q5 . . . . . Q 15 · A 16
10. Worksheet · Q3 . . . . . Q 15 · A 16
11. Worksheet · Q4 . . . . . Q 17 · A 18
12. Worksheet · Q5 . . . . . Q 17 · A 18
13. Worksheet · Q6 . . . . . Q 19 · A 20
14. Worksheet · Q7 . . . . . Q 19 · A 20
15. Kognity A · Q1 . . . . . Q 21 · A 22
16. Kognity A · Q4 . . . . . Q 23 · A 24
17. Kognity C · Q2 . . . . . Q 23 · A 24
18. Kognity D · Q1 . . . . . Q 23 · A 24
19. Kognity D · Q2 . . . . . Q 25 · A 26
20. Kognity D · Q4 . . . . . Q 25 · A 26
21. Kognity D · Q5 . . . . . Q 25 · A 26
22. Y13 Worksheet · Q7 . . . . . Q 25 · A 26
23. Y13 Worksheet · Q10 . . . . . Q 27 · A 28
24. Y13 Worksheet · Q11 . . . . . Q 27 · A 28
25. Y13 Worksheet · Q12 . . . . . Q 27 · A 28
26. Oxford Ex 14J · Q1 . . . . . Q 27 · A 28
27. Oxford Ex 14J · Q2 . . . . . Q 29 · A 30
28. Oxford Ex 14J · Q3 . . . . . Q 29 · A 30
29. Oxford Ex 14J · Q4 . . . . . Q 29 · A 30
30. Oxford Ex 14J · Q5 . . . . . Q 31 · A 32
31. SME Medium · Q2 . . . . . Q 33 · A 34
32. SME Medium · Q3 . . . . . Q 33 · A 34
33. SME Medium · Q6 . . . . . Q 33 · A 34
34. SME Hard · Q1 . . . . . Q 35 · A 36
35. SME Hard · Q4 . . . . . Q 35 · A 36
36. SME Hard · Q6 . . . . . Q 37 · A 38
37. SME Very Hard · Q4 . . . . . Q 39 · A 40

3. Inverse Normal & Unknown μ or σ

38. RV · Q14 . . . . . Q 41 · A 42
39. RV · Q17 . . . . . Q 41 · A 42
40. Worksheet · Q9 . . . . . Q 43 · A 44
41. Worksheet · Q10 . . . . . Q 43 · A 44
42. Worksheet · Q11 . . . . . Q 45 · A 46
43. Worksheet · Q12 . . . . . Q 47 · A 48
44. Worksheet · Q13 . . . . . Q 47 · A 48
45. Worksheet · Q14 . . . . . Q 49 · A 50
46. Worksheet · Q15 . . . . . Q 49 · A 50

4. Normal with Binomial, Conditional & Expected Numbers

47. RV · Q10 ..... Q 51 · A 52
48. RV · Q23 ..... Q 51 · A 52
49. RV · Q26 ..... Q 53 · A 54
50. RV · Q19 ..... Q 55 · A 56
51. RV · Q36 ..... Q 57 · A 58
52. Worksheet · Q16 ..... Q 59 · A 60
53. Worksheet · Q17 ..... Q 61 · A 62
54. Worksheet · Q18 ..... Q 63 · A 64
55. Worksheet · Q19 ..... Q 65 · A 66
56. Oxford Exam-style · Q21 ..... Q 67 · A 68
57. Oxford Exam-style · Q25 ..... Q 67 · A 68
58. SME Hard · Q10 ..... Q 69 · A 70
59. SME Very Hard · Q8 ..... Q 69 · A 70
60. SME Very Hard · Q10 ..... Q 71 · A 72

Key Notes

↑ back to Index

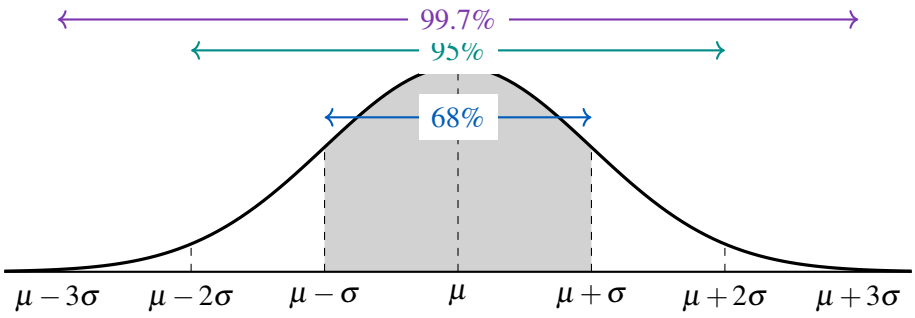
1. What is a normal distribution? — a CONTINUOUS distribution

A continuous random variable (heights, weights, times…) whose graph is a bell-shaped curve. $X \sim \mathrm{N}(\mu, \sigma^2)$: μ = mean, σ^2 = variance. **Careful:** $\mathrm{N}(20, 9)$ means $\sigma = 3$.

- Symmetric about μ: mean = median = mode; $\mathrm{P}(X < \mu) = \mathrm{P}(X > \mu) = 0.5$; total area $= 1$.
- Probability = area under the curve; $\mathrm{P}(X = a) = 0$, so $\mathrm{P}(X < a) = \mathrm{P}(X \leq a)$.

★ **Use boundary values exactly.** "Between 80 and 86" → `normCdf(80,86,μ,σ)`; "more than 80" → lower bound **80**, not 81. (Only the *binomial* uses 81.)

2. The 68–95–99.7 rule (approximately)



3. Standardising — the z-score

$$z = \frac{x - \mu}{\sigma} \quad \Longleftrightarrow \quad x = \mu + z\sigma, \qquad Z \sim \mathrm{N}(0, 1^2)$$

z = number of standard deviations above ($z > 0$) or below ($z < 0$) the mean. e.g. $\mu = 100$, $\sigma = 10$: 1.12 s.d. above is 111.2.

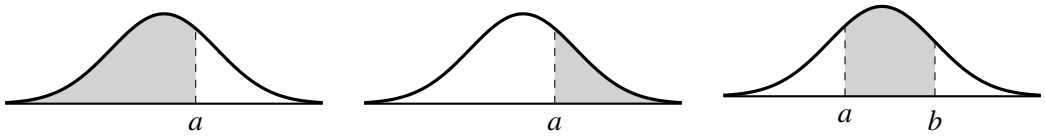
4. Using the TI-Nspire CX II

Know x, want a probability — menu $5 \to 5 \to$ **2: Normal Cdf**: $\mathrm{P}(a < X < b) =$ `normCdf(a,b,μ,σ)`; use $-$`9E99` / `9E99` for $\mp\infty$

Know a probability, want x — menu $5 \to 5 \to$ **3: Inverse Normal**: $x =$ `invNorm(left area,μ,σ)`

Always state $X \sim \mathrm{N}(\mu, \sigma^2)$ and write the probability statement before the answer.

5. Always sketch!



$\mathrm{P}(X < a)$ $\quad$ $\mathrm{P}(X > a) = 1 - \mathrm{P}(X < a)$ $\quad$ $\mathrm{P}(a < X < b)$

6. Using symmetry (no calculator needed)

- $\mathrm{P}(X < \mu - k) = \mathrm{P}(X > \mu + k)$: two tails the same distance from μ are equal.
- If $\mathrm{P}(X < d) = \mathrm{P}(X > 105)$ and $\mu = 100$, then $d = 95$.
- $\mathrm{P}(\mu < X < \mu + k) = 0.5 - \mathrm{P}(X > \mu + k)$; e.g. $\mathrm{P}(X > 24) = 0.06$, $\mu = 17 \Rightarrow \mathrm{P}(17 < X < 24) = 0.44$.

7. Inverse normal — always use the LEFT tail

★ **The TI-Nspire CX II Inverse Normal works with the area to the LEFT of x.** If a right-tail area is given, change it first: $\mathrm{P}(X > a) = 0.27 \Rightarrow \mathrm{P}(X < a) = 0.73 \Rightarrow a =$ `invNorm(0.73,μ,σ)`.

8. Finding an unknown mean μ or standard deviation σ

1. $z =$ `invNorm(left area, 0, 1)`.
2. Substitute into $z = \dfrac{x-\mu}{\sigma}$ and solve. e.g. $\sigma = 5$, $\mathrm{P}(X < 153) = 0.705$: $z = 0.5388 \Rightarrow \mu = 150$.

Both unknown: two facts give $x_1 = \mu + z_1\sigma$, $x_2 = \mu + z_2\sigma$ — solve simultaneously.

9. Normal + binomial together; expected numbers

First use the normal distribution to find p for one item; then for n items $Y \sim \mathrm{B}(n, p)$. e.g. all 10 chickens: p^{10}; at least 7: `binomCdf(10,p,7,10)`; two items both in a range: p^2.

★ **Expected numbers are not rounded**: $\mathrm{E} = n \times p$, e.g. $450 \times 0.8176 = 367.9$ frogs (keep the decimal) — unless the question asks for an integer.

10. Conditional probability (tree diagrams)

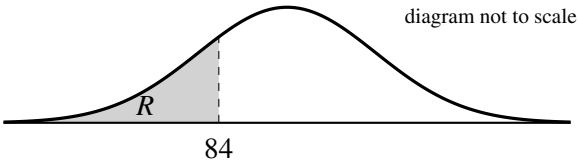
$$\mathrm{P}(A \,|\, B) = \frac{\mathrm{P}(A \cap B)}{\mathrm{P}(B)} \qquad \text{e.g.} \quad \mathrm{P}(\text{girl} \,|\, \text{tall}) = \frac{\mathrm{P}(\text{girl}) \times \mathrm{P}(\text{tall} \,|\, \text{girl})}{\mathrm{P}(\text{tall})}$$

For one variable: $\mathrm{P}(X > 55 \,|\, X > 48) = \dfrac{\mathrm{P}(X > 55)}{\mathrm{P}(X > 48)}$.

1 answer → No calculator **RV · Q2**

[Maximum mark: 5]

The random variable X is normally distributed with a mean of 90. The following diagram shows the normal curve for X.



Let R be the shaded region under the curve, to the left of 84. The area of R is 0.22.

(a) Write down $\mathrm{P}(X < 84)$. [1]

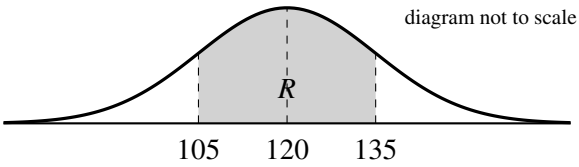
(b) Find $\mathrm{P}(84 < X < 90)$. [2]

(c) Find $\mathrm{P}(84 < X < 96)$. [2]

2 answer → No calculator **RV · Q3**

[Maximum mark: 5]

The random variable X is normally distributed with a mean of 120. The following diagram shows the normal curve for X.



Let R be the shaded region under the curve between 105 and 135. The area of R is 0.4.

(a) Write down $\mathrm{P}(105 < X < 135)$. [1]

(b) Find $\mathrm{P}(X < 135)$. [2]

(c) Find $\mathrm{P}(X > 105 \,|\, X < 135)$. [2]

1 ← question *RV · Q2*

(a) $\boxed{0.22}$ **A1**

(b) $0.5 - 0.22 = \boxed{0.28}$ **M1 A1**

(c) 96 is also 6 from the mean: $2 \times 0.28 = \boxed{0.56}$ **M1 A1**

2 ← question *RV · Q3*

(a) $\boxed{0.4}$ **A1**

(b) $\mathrm{P}(X < 120) + \dfrac{R}{2} = 0.5 + 0.2 = \boxed{0.7}$ **M1 A1**

(c) $\dfrac{\mathrm{P}(105 < X < 135)}{\mathrm{P}(X < 135)} = \dfrac{0.4}{0.7} = \boxed{\dfrac{4}{7}}$ **M1 A1**

3 answer → No calculator **Worksheet · Q1**

[Maximum mark: 6]

A random variable X is distributed normally with a mean of 100 and a variance of 100.

(a) Find the value of X that is 1.12 standard deviations above the mean. [4]

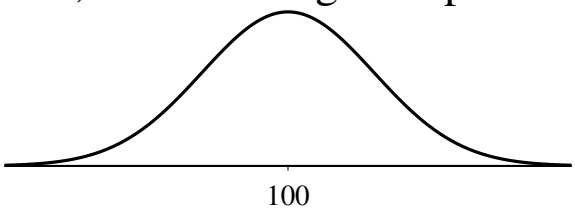
(b) Find the value of X that is 1.12 standard deviations below the mean. [2]

4 answer → No calculator **Worksheet · Q2**

[Maximum mark: 6]

Let X be normally distributed with mean 100 cm and standard deviation 5 cm.

(a) On the diagram below, shade the region representing $P(X > 105)$. [2]



(b) Given that $P(X < d) = P(X > 105)$, find the value of d. [2]

(c) Given that $P(X > 105) = 0.16$ (correct to two significant figures), find $P(d < X < 105)$. [2]

3 ← question *Worksheet · Q1*

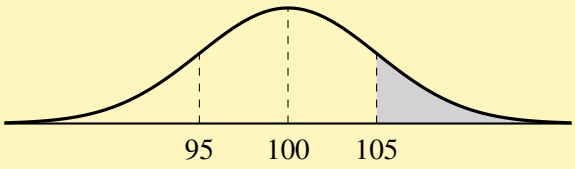
Variance $= 100 \Rightarrow \sigma = 10$ **A1**

(a) $x = \mu + 1.12\sigma = 100 + 11.2 = \boxed{111.2}$ **M1 A2**

(b) $x = 100 - 11.2 = \boxed{88.8}$ **A2**

4 ← question *Worksheet · Q2*

(a)



A2

(b) by symmetry $d = 100 - 5 = \boxed{95}$ **A2**

(c) $1 - 0.16 - 0.16 = \boxed{0.68}$ **M1 A1**

5 answer →

No calculator **Worksheet · Q8**

[Maximum mark: 9]

The heights of trees in a forest are normally distributed with mean height 17 metres. One tree is selected at random. The probability that a selected tree has a height greater than 24 metres is 0.06.

(a) Find the probability that the tree selected has a height less than 24 metres. [2]

(b) The probability that the tree has a height less than D metres is 0.06. Find the value of D. [3]

(c) A woodcutter randomly selects 200 trees. Find the expected number of trees whose height lies between 17 metres and 24 metres. [4]

6 answer →

No calculator **Kognity B · Q3**

[Maximum mark: 1]

Let $Z \sim \mathrm{N}(0,1)$ and $0.03 < \mathrm{P}(Z > c) < 0.12$. Which of the following is true about c?

1. $2 < c < 3$ **2.** $-2 < c < -1$ **3.** $1 < c < 2$ **4.** $-3 < c < -2$

7 answer →

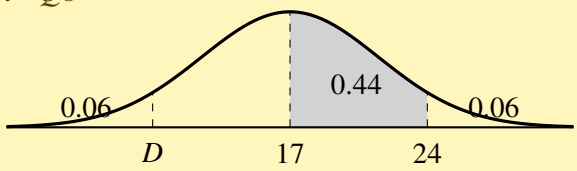
No calculator **Kognity B · Q4**

[Maximum mark: 1]

Fiona discovers that the lifetime of washing machines is normally distributed with mean 8 years and standard deviation 2 years. What is the probability of a machine lasting between 6 and 10 years?

1. 68% **2.** 50% **3.** 10% **4.** 25%

5 ← question *Worksheet · Q8*



(a) $1 - 0.06 = \boxed{0.94}$ **A2**

(b) 24 is 7 above the mean, so $D = 17 - 7 = \boxed{10}$ **M1 A2**

(c) $\mathrm{P}(17 < H < 24) = 0.5 - 0.06 = 0.44$; $200 \times 0.44 = \boxed{88}$ trees **M1**

A1 M1 A1

6 ← question *Kognity B · Q3*

68–95–99.7: $\mathrm{P}(Z > 1) \approx 0.16$, $\mathrm{P}(Z > 2) \approx 0.025$. $0.03 < \mathrm{P}(Z > c) < 0.12$ lies between these, so $\boxed{1 < c < 2}$ (Option 3) **A1**

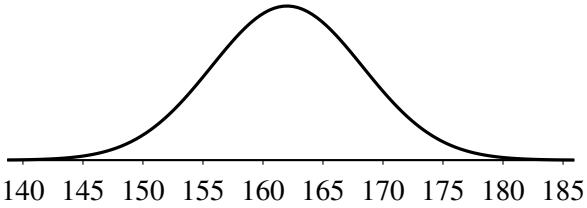
7 ← question *Kognity B · Q4*

6 and 10 are $\mu \pm \sigma$, so about $\boxed{68\%}$ (Option 1) **A1**

8 answer → Calculator **SME Medium · Q1**

[Maximum mark: 9]

The random variable X shows the distribution of heights, in cm, of adult women in the UK. The distribution of heights follows a normal distribution, with a mean of 162 cm and a standard deviation of 6.3 cm.



(a) On the diagram above, shade in the region representing $P(X > 155)$. [2]

(b) (i) Find the probability that a randomly selected woman has a height of more than 155 cm. (ii) Use your answer from part (b)(i) to find the probability that a randomly selected woman has a height of more than 169 cm. [4]

(c) Suggest a range of heights within which the height of approximately (i) 68% (ii) 95% (iii) 99.7% of adult women in the UK will fall. [3]

8 ← question *SME Medium · Q1*

(a) shade to the right of 155 (just left of the mean) **A2**

(b)(i) $\mathrm{P}(X > 155) = \boxed{0.867}$ **M1 A1**

(ii) 169 is 7 above, 155 is 7 below: $1 - 0.867 = \boxed{0.133}$ **M1 A1**

(c) (i) 162 ± 6.3: $\boxed{155.7 \text{ to } 168.3}$ (ii) $\boxed{149.4 \text{ to } 174.6}$ (iii) $\boxed{143.1 \text{ to } 180.9}$

A1 A1 A1

TI-Nspire CX: Question 8

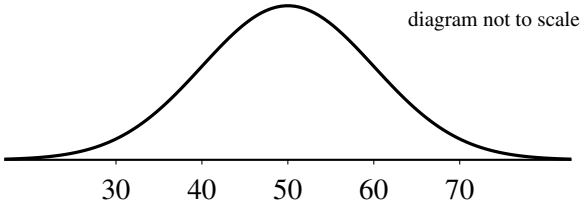
menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use $\pm$`9E99` for $\pm\infty$). `normCdf(155,9E99,162,6.3)`

9 answer → Calculator **RV · Q5**

[Maximum mark: 6]

A random variable X is normally distributed with a mean of 50 and a standard deviation of 10.

(a) On the following diagram, shade in the region representing $P(X \geq 45)$. [2]



(b) Find $P(X \geq 45)$. [2]

(c) Given that $P(X \leq k) = 0.8$, find the value of k. [2]

10 answer → Calculator **Worksheet · Q3**

[Maximum mark: 6]

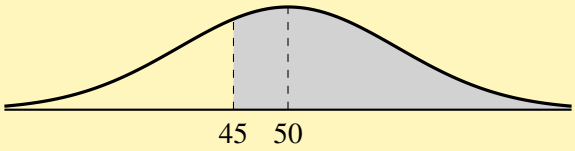
A random variable X is distributed normally with mean 450 and standard deviation 20.

(a) Find $P(X \leq 475)$. [2]

(b) Given that $P(X > a) = 0.27$, find a. [4]

9 ← question *RV · Q5*

(a)



A2(b) $\boxed{0.691}$ **M1 A1**

(c) $k = \boxed{58.4}$ **M1 A1**

TI-Nspire CX: Question 9

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(45,9E99,50,10)`; `invNorm(0.8,50,10)`

10 ← question *Worksheet · Q3*

(a) $\boxed{0.894}$ **M1 A1**

(b) right tail → left tail: $\mathrm{P}(X < a) = 0.73$ **M1 A1**

$a = \boxed{462}$ (462.256) **A2**

TI-Nspire CX: Question 10

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(-9E99,475,450,20)`; `invNorm(0.73,450,20)`

11 answer → Calculator **Worksheet · Q4**

[Maximum mark: 6]

The heights of a group of students are normally distributed with a mean of 160 cm and a standard deviation of 20 cm.

(a) A student is chosen at random. Find the probability that the student's height is greater than 180 cm. [3]

(b) In this group of students, 11.9% have heights less than d cm. Find the value of d. [3]

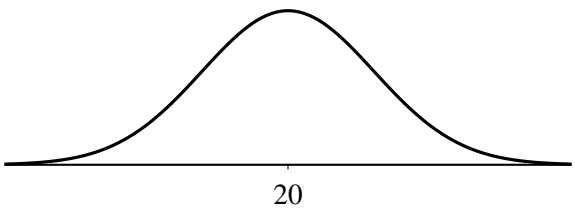
12 answer → Calculator **Worksheet · Q5**

[Maximum mark: 8]

A random variable X is distributed normally with a mean of 20 and variance 9.

(a) Find $\mathrm{P}(X \leq 24.5)$. [3]

(b) Let $\mathrm{P}(X \leq k) = 0.85$. (i) Represent this information on the following diagram.



(ii) Find the value of k. [5]

11 ← question *Worksheet · Q4*

(a) $\mathrm{P}(H > 180) = \boxed{0.159}$ (180 is exactly 1σ above) **M1 A2**

(b) $d = \text{invNorm}(0.119, 160, 20) = \boxed{136}$ **M1 A2**

TI-Nspire CX: Question 11

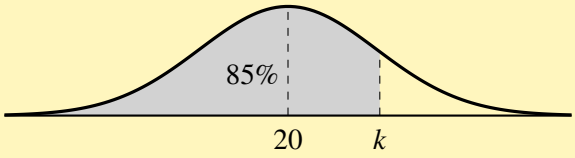
menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use $\pm$9E99 for $\pm\infty$). `normCdf(180,9E99,160,20); invNorm(0.119,160,20)`

12 ← question *Worksheet · Q5*

$\sigma = \sqrt{9} = 3$ **A1**

(a) $\boxed{0.933}$ **M1 A1**

(b)(i)



A1(ii) $k = \boxed{23.1}$ **M1 A1**

TI-Nspire CX: Question 12

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use $\pm$9E99 for $\pm\infty$). `normCdf(-9E99,24.5,20,3); invNorm(0.85,20,3)`

13 answer → Calculator **Worksheet · Q6**

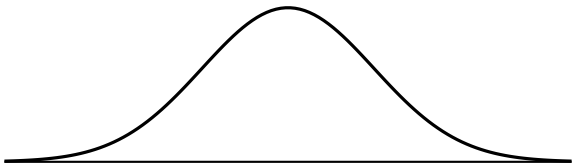
[Maximum mark: 6]

The weights of a group of children are normally distributed with a mean of 22.5 kg and a standard deviation of 2.2 kg.

(a) Write down the probability that a child selected at random has a weight more than 25.8 kg. [2]

(b) Of the group 95% weigh less than k kilograms. Find the value of k. [2]

(c) The diagram below shows a normal curve. On the diagram, shade the region that represents the following information: 87% of the children weigh less than 25 kg. [2]



14 answer → Calculator **Worksheet · Q7**

[Maximum mark: 8]

Residents of a small town have savings which are normally distributed with a mean of \$3000 and a standard deviation of \$500.

(i) What percentage of townspeople have savings greater than \$3200?

(ii) Two townspeople are chosen at random. What is the probability that both of them have savings between \$2300 and \$3300?

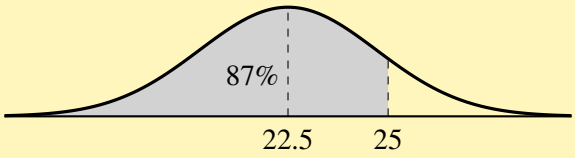
(iii) The percentage of townspeople with savings less than d dollars is 74.22%. Find the value of d.

13 ← question *Worksheet · Q6*

(a) $\boxed{0.0668}$ **A2**

(b) $k = \boxed{26.1}$ kg **M1 A1**

(c)



A2

TI-Nspire CX: Question 13

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(25.8,9E99,22.5,2.2)`; `invNorm(0.95,22.5,2.2)`

14 ← question *Worksheet · Q7*

(i) $0.345 \Rightarrow \boxed{34.5\%}$ **M1 A1**

(ii) $0.645^2 = \boxed{0.416}$ **M1 A1 M1 A1**

(iii) $d = \$\boxed{3325}$ ($3330 to 3 s.f.) **M1 A1**

TI-Nspire CX: Question 14

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(3200,9E99,3000,500)`; `normCdf(2300,3300,3000,500)^2`; `invNorm(0.7422,3000,500)`

15 answer → Calculator **Kognity A · Q1**

[Maximum mark: 2]

In US dollars, the 100 movies that grossed the most money worldwide had a mean $\mu = 491$ million with a standard deviation $\sigma = 379$ million. Assuming the amounts are normally distributed, find the probability that a movie grossed more than \$870 million. Give your answer as a decimal to two decimal places.

15 ← question *Kognity A · Q1*

870 is exactly 1σ above the mean: $\mathrm{P}(X > 870) = 0.1587 \approx \boxed{0.16}$ **M1**

A1

TI-Nspire CX: Question 15

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(870,9E99,491,379)`

16 answer → Calculator **Kognity A · Q4**

[Maximum mark: 4]

A soft drink bottling company fills 2-litre bottles, but the bottle-filling machine is not perfect. The volume with which the bottles are filled is distributed normally with $\mu = 2000$ millilitres and $\sigma = 1.5$ millilitres.

(a) Find the probability that a bottle is filled between 1998 and 2004 millilitres. Give your answer as a decimal rounded to 3 significant figures. [2]

(b) A bottle is considered underfilled if it contains less than 1997 millilitres. Find the probability that a randomly selected bottle is **not** underfilled. Give your answer to 3 decimal places. [2]

17 answer → Calculator **Kognity C · Q2**

[Maximum mark: 2]

A company manufactures steel tubes with an average circumference of 12.5 cm. The maximum divergence allowed is 0.15 cm. The circumference of the tubes follows a normal distribution with standard deviation 0.09 cm. What percentage of tubes have an unacceptably large circumference?

1. 4.78% **2.** 4.50% **3.** 5.25% **4.** 5.00%

18 answer → Calculator **Kognity D · Q1**

[Maximum mark: 2]

The marks in a maths test are normally distributed with mean 66 and standard deviation 9. What is the probability that a student achieves more than 90 marks in the test? Give your answer correct to 3 significant figures.

16 ← question *Kognity A · Q4*

(a) $P(1998 < X < 2004) = \boxed{0.905}$ **M1 A1**

(b) $P(X \geq 1997) = \boxed{0.977}$ **M1 A1**

(Parts (c) and (d) of this question are in the Inverse Normal booklet.)

TI-Nspire CX: Question 16

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(1998,2004,2000,1.5); normCdf(1997,9E99,2000,1.5)`

17 ← question *Kognity C · Q2*

Too large: $C > 12.5 + 0.15 = 12.65$ **M1**

$P(C > 12.65) = 0.0478 = \boxed{4.78\%}$ (Option 1) **A1**

TI-Nspire CX: Question 17

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(12.65,9E99,12.5,0.09)`

18 ← question *Kognity D · Q1*

$P(X > 90) = \boxed{0.00383}$ **M1 A1**

TI-Nspire CX: Question 18

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(90,9E99,66,9)`

19 answer → Calculator **Kognity D · Q2**

[Maximum mark: 2]

The weights of potatoes in a bag are normally distributed with mean 185 g and standard deviation 34 g. What is the probability that a potato selected at random weighs less than 150 g? Give your answer correct to 3 significant figures.

20 answer → Calculator **Kognity D · Q4**

[Maximum mark: 2]

The weights of newborn giraffes are normally distributed with mean 55 kg and standard deviation 10 kg. What percentage of giraffes selected at random weigh between 50 kg and 60 kg? Give your answer correct to 3 significant figures.

21 answer → Calculator **Kognity D · Q5**

[Maximum mark: 2]

In an albatross colony the wingspan of the birds is approximately normally distributed with an average wingspan of 3.07 metres and standard deviation 0.48 metres. What percentage of the birds have wingspan between 2.91 and 3.59 metres? Give your answer rounded to the nearest integer.

22 answer → Calculator **Y13 Worksheet · Q7**

[Maximum mark: 2]

The weights of apples are normally distributed with mean 120 g and standard deviation 10 g. Find the probability that an apple weighs between 115 g and 130 g.

19 ← question *Kognity D · Q2*

$\mathrm{P}(X < 150) = \boxed{0.152}$ **M1 A1**

TI-Nspire CX: Question 19

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(-9E99,150,185,34)`

20 ← question *Kognity D · Q4*

$\mathrm{P}(50 < X < 60) = 0.383 = \boxed{38.3\%}$ **M1 A1**

TI-Nspire CX: Question 20

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(50,60,55,10)`

21 ← question *Kognity D · Q5*

$\mathrm{P}(2.91 < X < 3.59) = 0.491 \Rightarrow \boxed{49}$ **M1 A1**

TI-Nspire CX: Question 21

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(2.91,3.59,3.07,0.48)`

22 ← question *Y13 Worksheet · Q7*

$z_1 = -0.5,\ z_2 = 1:\ \ 0.8413 - 0.3085 = \boxed{0.533}$ **M1 A1**

TI-Nspire CX: Question 22

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(115,130,120,10)`

23 answer → Calculator **Y13 Worksheet · Q10**

[Maximum mark: 2]

A random variable $X \sim \mathrm{N}(40, 9)$. Find $\mathrm{P}(X > 44.5)$.

24 answer → Calculator **Y13 Worksheet · Q11**

[Maximum mark: 2]

The time taken to assemble a part is normally distributed with mean 50 seconds and $\sigma = 4$ seconds. Find the probability that a randomly chosen part takes less than 45 seconds.

25 answer → Calculator **Y13 Worksheet · Q12**

[Maximum mark: 3]

A machine fills packets with a mean of 200 g and standard deviation 3 g. Find the proportion of packets weighing between 194 g and 206 g. Hence find the probability a randomly chosen packet is within 6 g of the mean.

26 answer → Calculator **Oxford Ex 14J · Q1**

[Maximum mark: 6]

Households in Portugal spend a mean of €100 per week on groceries with a standard deviation of €20. Assuming that the distribution of grocery expenditure follows a normal distribution, find the probability of a household spending:

(a) less than €130 per week [2]

(b) more than €90 per week [2]

(c) between €80 and €125 per week. [2]

23 ← question *Y13 Worksheet · Q10*

$\sigma = \sqrt{9} = 3$ (variance is 9) **M1**

$P(X > 44.5) = \boxed{0.0668}$ **A1**

TI-Nspire CX: Question 23

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(44.5,9E99,40,3)`

24 ← question *Y13 Worksheet · Q11*

$P(X < 45) = \boxed{0.106}$ $(z = -1.25)$ **M1 A1**

TI-Nspire CX: Question 24

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(-9E99,45,50,4)`

25 ← question *Y13 Worksheet · Q12*

$P(194 < X < 206) = \boxed{0.954}$ $(z = \pm 2)$ **M1 A1**

within 6 g of the mean is the same event: $\boxed{0.954}$ (95.4%) **A1**

TI-Nspire CX: Question 25

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for ±∞). `normCdf(194,206,200,3)`

26 ← question *Oxford Ex 14J · Q1*

$X \sim N(100, 20^2)$

(a) $\boxed{0.933}$ (b) $\boxed{0.691}$ (c) $\boxed{0.736}$ **A2 A2 A2**

TI-Nspire CX: Question 26

`normCdf(-9E99,130,100,20)`, `normCdf(90,9E99,100,20)`, `normCdf(80,125,100,20)`

27 answer → Calculator **Oxford Ex 14J · Q2**

[Maximum mark: 4]

A machine produces bolts with diameters that are normally distributed with a mean of 4 mm and a standard deviation of 0.25 mm. Bolts are measured accurately, and any bolts which are smaller than 3.5 mm or bigger than 4.5 mm are rejected. Out of a batch of 500 bolts, find the number of bolts you would expect to be accepted.

28 answer → Calculator **Oxford Ex 14J · Q3**

[Maximum mark: 4]

The length of time patients have to wait in Dr Barrett's waiting room is known to be normally distributed with a mean of 14 minutes and a standard deviation of 4 minutes.

(a) Find the probability that a patient will have to wait more than 20 minutes to see Dr Barrett. [2]

(b) Find the percentage of patients who wait for less than 10 minutes. [2]

29 answer → Calculator **Oxford Ex 14J · Q4**

[Maximum mark: 2]

A company that produces breakfast cereal claims that packets of "flakey flakes" have a net mass of 550 g. The production of packets is such that the masses are normally distributed with a mean of 551.3 g and standard deviation of 15 g. Find the percentage of packets which are heavier than the stated mass.

27 ← question *Oxford Ex 14J · Q2*

$P(3.5 < X < 4.5) = 0.9545$ **M1 A1**

$500 \times 0.9545 = \boxed{477.2}$ bolts (do not round) **M1 A1**

TI-Nspire CX: Question 27

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(3.5,4.5,4,0.25)`

28 ← question *Oxford Ex 14J · Q3*

(a) $\boxed{0.0668}$ **M1 A1**

(b) $0.1587 \Rightarrow \boxed{15.9\%}$ **M1 A1**

TI-Nspire CX: Question 28

`normCdf(20,9E99,14,4); normCdf(-9E99,10,14,4)`

29 ← question *Oxford Ex 14J · Q4*

$P(X > 550) = 0.535 \Rightarrow \boxed{53.5\%}$ **M1 A1**

TI-Nspire CX: Question 29

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(550,9E99,551.3,15)`

30 answer → Calculator **Oxford Ex 14J · Q5**

[Maximum mark: 4]

The masses of packets of washing powder are normally distributed with a mean of 500 g and standard deviation of 20 g.

(a) Find the probability that a packet chosen at random has a mass less than 475 g. [2]

(b) Three packets are chosen at random. Determine the probability that all three packets have masses less than 475 g. [2]

30 ← question *Oxford Ex 14J · Q5*

(a) $\boxed{0.106}$ **M1 A1**

(b) $0.105650^3 = \boxed{0.00118}$ **M1 A1**

TI-Nspire CX: Question 30

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use $\pm$`9E99` for $\pm\infty$). `normCdf(-9E99,475,500,20)`

31 answer → Calculator SME Medium · Q2

[Maximum mark: 6]

(a) For the random variable $X \sim \mathrm{N}(23, 4^2)$ find: (i) $\mathrm{P}(X < 20)$ (ii) $\mathrm{P}(X \geq 29)$ (iii) $\mathrm{P}(20 \leq X < 29)$ [3]

(b) For the random variable $Y \sim \mathrm{N}(100, 225)$ find: (i) $\mathrm{P}(Y \leq 90)$ (ii) $\mathrm{P}(Y > 140)$ (iii) $\mathrm{P}(85 \leq Y \leq 115)$ [3]

32 answer → Calculator SME Medium · Q3

[Maximum mark: 4]

(a) The weight, W g, of a chocolate bar produced by a certain manufacturer is modelled as $W \sim \mathrm{N}(200, 1.75^2)$. Find: (i) $\mathrm{P}(W < 195)$ (ii) $\mathrm{P}(W > 203)$ [2]

(b) Heledd buys a pack containing 12 of the chocolate bars. It may be assumed that the 12 bars in the pack represent a random sample. Find the probability that all of the bars in the pack have a weight of at least 195 g. [2]

33 answer → Calculator SME Medium · Q6

[Maximum mark: 7]

(a) For the standard normal distribution $Z \sim \mathrm{N}(0, 1^2)$, find: (i) $\mathrm{P}(Z < 1.5)$ (ii) $\mathrm{P}(Z > -0.8)$ (iii) $\mathrm{P}(-2.1 < Z < -0.3)$ [4]

(b) The random variable $X \sim \mathrm{N}(2, 0.1^2)$. By using the coding relationship between Z and X, re-express the probabilities from parts (a)(i), (ii) and (iii) in the forms $\mathrm{P}(X < a)$, $\mathrm{P}(X > b)$ and $\mathrm{P}(c < X < d)$ respectively, where a, b, c and d are constants to be found. [3]

31 ← question *SME Medium · Q2*

(a)(i) $\boxed{0.227}$ (ii) $\boxed{0.0668}$ (iii) $\boxed{0.707}$ **A1 A1 A1**

(b) $\sigma = \sqrt{225} = 15$: (i) $\boxed{0.252}$ (ii) $\boxed{0.00383}$ (iii) $\boxed{0.683}$ **A1 A1 A1**

TI-Nspire CX: Question 31

Boundaries are used exactly ($\leq$ and $<$ give the same area). e.g. `normCdf(20,29,23,4)`, `normCdf(85,115,100,15)`

32 ← question *SME Medium · Q3*

(a)(i) $\boxed{0.00214}$ (ii) $\boxed{0.0432}$ **A1 A1**

(b) $(1-0.002137)^{12} = \boxed{0.975}$ **M1 A1**

TI-Nspire CX: Question 32

`normCdf(-9E99,195,200,1.75); (1-ans)^12`

33 ← question *SME Medium · Q6*

(a)(i) $\boxed{0.933}$ (ii) $\boxed{0.788}$ (iii) $\boxed{0.364}$ **A1 A1 A2**

(b) $x = 2+0.1z$: $P(X < \boxed{2.15})$, $P(X > \boxed{1.92})$, $P(\boxed{1.79} < X < \boxed{1.97})$ **A1 A1 A1**

TI-Nspire CX: Question 33

`normCdf(-2.1,-0.3,0,1)` etc.

34 answer → Calculator **SME Hard · Q1**

[Maximum mark: 6]

(a) A scientist is studying the movement of snails and has observed that the distribution of their speeds, S, follows a normal distribution with a mean of 48 m/h and a standard deviation of 1.5 m/h. Sketch a diagram to represent this information. [2]

(b) Find the probability that a randomly selected snail has a speed of less than 46.5 m/h. [2]

(c) From a sample of 80 snails, calculate the expected number of snails that would have a speed of less than 46.5 m/h. Give your answer to the nearest integer. [2]

35 answer → Calculator **SME Hard · Q4**

[Maximum mark: 6]

(a) The weight, W, of pumpkins purchased are normally distributed with a mean of 11.3 kg and a standard deviation of 2.1 kg. In one year, a farmer grows 350 pumpkins on his farm. Predict the number of pumpkins that weigh between 7.2 kg and 12.5 kg from the farm. [3]

(b) The heaviest 7% of pumpkins are classified as large and are sold at a higher price. Find the range of weights of pumpkins that can be sold for a higher price. [3]

34 ← question *SME Hard · Q1*

(a) bell curve centred at 48, marks at $46.5, 49.5$ ($\pm 1\sigma$), $45, 51$ ($\pm 2\sigma$) **A2**

(b) $\mathrm{P}(S < 46.5) = \boxed{0.159}$ **M1 A1**

(c) $80 \times 0.1587 = 12.69 \approx \boxed{13}$ (the question asks for an integer) **M1 A1**

TI-Nspire CX: Question 34

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(-9E99,46.5,48,1.5)`

35 ← question *SME Hard · Q4*

(a) $\mathrm{P}(7.2 < W < 12.5) = 0.69070$; $350 \times 0.69070 = \boxed{241.7}$ pumpkins **M1 A1 A1**

(b) $\mathrm{P}(W < w) = 0.93$: $w = 14.4$; $\boxed{W > 14.4 \text{ kg}}$ **M1 A1 A1**

TI-Nspire CX: Question 35

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(7.2,12.5,11.3,2.1); invNorm(0.93,11.3,2.1)`

36 answer → Calculator **SME Hard · Q6**

[Maximum mark: 7]

(a) A reaction time test has been used to collate data about how quickly a person can react to a given signal. It is found that the results of the test, X, are distributed normally with a mean of 273 milliseconds and a variance of 121 milliseconds. Use a sketch of the distribution to show that $P(284 < X < 295) = 0.135$. [3]

(b) A result that lies outside of two and a half standard deviations from the mean is considered to be extreme. A group of 145 students decide to measure their reaction times using the test. Estimate the number of students that will receive a result that would be considered extreme. Give your answer to the nearest integer. [4]

36 ← question *SME Hard · Q6*

(a) $\sigma = 11$: 284 and 295 are $\mu+\sigma$ and $\mu+2\sigma$; area $\approx \dfrac{95\%-68\%}{2} =$ 13.5% ✓ (GDC: 0.1359) **M1 A1 R1**

(b) $\mathrm{P}(|Z| > 2.5) = 0.012419$; $145 \times 0.012419 = 1.80 \approx \boxed{2}$ students **M1 A1 M1 A1**

TI-Nspire CX: Question 36

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(284,295,273,11)`

37 answer → Calculator **SME Very Hard · Q4**

[Maximum mark: 8]

The life expectancy, L, in hours of two brands of vacuum cleaner is normally distributed: Brand A: $L \sim \mathrm{N}(1500, 273^2)$; Brand B: $L \sim \mathrm{N}(1450, 146^2)$.

(a) State which vacuum cleaner brand is more reliable. Give a reason for your answer. [2]

(b) Using the information for the brand you stated in part (a), find the probability that the life expectancy of the vacuum will be between 1550 hours and 1700 hours. [2]

(c) When sold, each brand B vacuum cleaner comes with a 5-year warranty. It is assumed that a typical customer uses the vacuum cleaner for approximately 4 hours each week. Find the probability that from a batch of 620 brand B vacuum cleaners, more than 0.5% of them could be returned to the manufacturer within the warranty period. [4]

37 ← question *SME Very Hard · Q4*

(a) **Brand B** — smaller standard deviation, so lifetimes are more consistent **A1 R1**

(b) $\mathrm{P}(1550 < L < 1700) = \boxed{0.203}$ **M1 A1**

(c) $5 \times 52 \times 4 = 1040$ hours; $p = \mathrm{P}(L < 1040) = 0.0024908$ **M1 A1**

0.5% of $620 = 3.1 \Rightarrow X \geq 4$; $X \sim \mathrm{B}(620, p)$: $\mathrm{P}(X \geq 4) = \boxed{0.0711}$ **M1 A1**

TI-Nspire CX: Question 37

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±9E99 for $\pm\infty$). `normCdf(1550,1700,1450,146); binomCdf(620,0.0024908,4,620)`

38 answer → Calculator **RV · Q14**

[Maximum mark: 6]

A random variable X is normally distributed with $\mu = 120$ and $\sigma = 15$. Find the interquartile range of X.

39 answer → Calculator **RV · Q17**

[Maximum mark: 7]

The time taken for a student to complete an exam is normally distributed with a mean of 40 minutes and a standard deviation of 5.5 minutes.

(a) A student is selected at random. Find the probability that the student completes the task in less than 48 minutes. [3]

(b) The probability a student takes between k and 48 minutes is 0.4. Find the value of k. [4]

38 ← question *RV · Q14*

$Q_3 = \text{invNorm}(0.75, 120, 15) = 130.117$ **M1 A1**

$Q_1 = \text{invNorm}(0.25, 120, 15) = 109.883$ **A1**

$\text{IQR} = \boxed{20.2}$ **M1 A2**

TI-Nspire CX: Question 38

`invNorm(0.75,120,15) − invNorm(0.25,120,15)`

39 ← question *RV · Q17*

(a) $\boxed{0.927}$ **M1 A2**

(b) $\text{P}(T < k) = 0.927102 - 0.4 = 0.527102$ **M1 A1**

$k = \boxed{40.4}$ minutes **M1 A1**

TI-Nspire CX: Question 39

menu → **5: Probability** → **5: Distributions** → **2: Normal Cdf**: Lower, Upper, μ, σ (use ±`9E99` for $\pm\infty$). `normCdf(-9E99,48,40,5.5); invNorm(0.527102,40,5.5)`

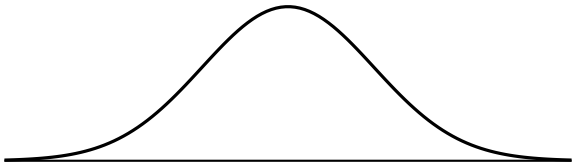
40 answer → Calculator **Worksheet · Q9**

[Maximum mark: 13]

A box contains a large number of biscuits. The weights of biscuits are normally distributed with mean 7 g and standard deviation 0.5 g.

(a) One biscuit is chosen at random from the box. Find the probability that this biscuit (i) weighs less than 8 g; (ii) weighs between 6 g and 8 g. [4]

(b) Five percent of the biscuits in the box weigh less than d grams. (i) Complete the following normal distribution diagram, to represent this information, by indicating d, and shading the appropriate region. (ii) Find the value of d. [5]



(c) The weights of biscuits in another box are normally distributed with mean μ and standard deviation 0.5 g. It is known that 20% of the biscuits in this second box weigh less than 5 g. Find the value of μ. [4]

41 answer → Calculator **Worksheet · Q10**

[Maximum mark: 6]

The heights of boys at a particular school follow a normal distribution with a standard deviation of 5 cm. The probability of a boy being shorter than 153 cm is 0.705.

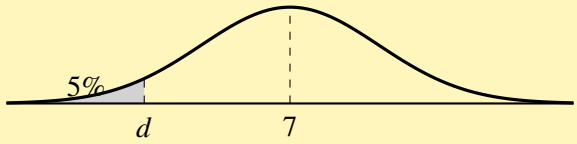
(a) Calculate the mean height of the boys. [3]

(b) Find the probability of a boy being taller than 156 cm. [3]

40 ← question *Worksheet · Q9*

(a)(i) $\boxed{0.977}$ (ii) $\boxed{0.954}$ **A2 A2**

(b)(i)



(ii) $d = \boxed{6.18}$ g **M1 A1**

(c) $z = -0.8416$; $\dfrac{5-\mu}{0.5} = -0.8416 \Rightarrow \mu = \boxed{5.42}$ g **M1 A1 M1 A1**

TI-Nspire CX: Question 40

```
normCdf(6,8,7,0.5); invNorm(0.05,7,0.5); invNorm(0.2,0,1)
```

41 ← question *Worksheet · Q10*

(a) $z = 0.5388$; $\dfrac{153-\mu}{5} = 0.5388 \Rightarrow \mu = \boxed{150}$ cm (150.31) **M1 A1 A1**

(b) $\mathrm{P}(H > 156) = \boxed{0.127}$ **M1 A2**

TI-Nspire CX: Question 41

```
invNorm(0.705,0,1); normCdf(156,9E99,150.306,5)
```

42 answer → Calculator **Worksheet · Q11**

[Maximum mark: 7]

The scores of a test given to students are normally distributed with a mean of 21. 80% of the students have scores less than 23.7.

(a) Find the standard deviation of the scores. [3]

(b) A student is chosen at random. This student has the same probability of having a score less than 25.4 as having a score greater than b. (i) Find the probability the student has a score less than 25.4. (ii) Find the value of b. [4]

42 ← question *Worksheet · Q11*

(a) $z = 0.8416$; $\sigma = \dfrac{2.7}{0.8416} = \boxed{3.21}$ **M1 A1 A1**

(b)(i) $\boxed{0.915}$ **M1 A1**

(ii) by symmetry $b = 21 - 4.4 = \boxed{16.6}$ **M1 A1**

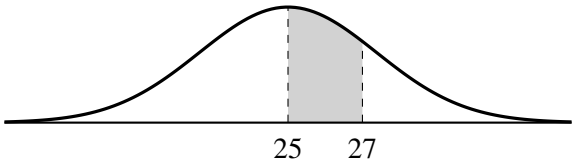
TI-Nspire CX: Question 42

```
invNorm(0.8,0,1); normCdf(-9E99,25.4,21,3.208)
```

43 answer → Calculator **Worksheet · Q12**

[Maximum mark: 7]

Let the random variable X be normally distributed with mean 25, as shown in the following diagram.



The shaded region between 25 and 27 represents 30% of the distribution.

(a) Find $P(X > 27)$. [2]

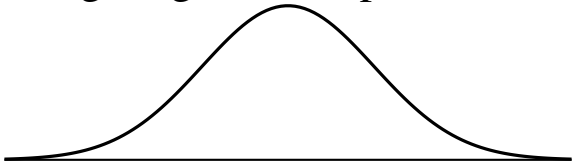
(b) Find the standard deviation of X. [5]

44 answer → Calculator **Worksheet · Q13**

[Maximum mark: 7]

The heights of certain plants are normally distributed. The plants are classified into three categories. The shortest 12.92% are in category A. The tallest 10.38% are in category C. All the other plants are in category B with heights between r cm and t cm.

(a) Complete the following diagram to represent this information. [2]



(b) Given that the mean height is 6.84 cm and the standard deviation 0.25 cm, find the value of r and of t. [5]

43 ← question *Worksheet · Q12*

(a) $0.5 - 0.3 = \boxed{0.2}$ **M1 A1**

(b) $\mathrm{P}(X < 27) = 0.8$, $z = 0.8416$; $\sigma = \dfrac{2}{0.8416} = \boxed{2.38}$ **M1 A1 M1 A1**

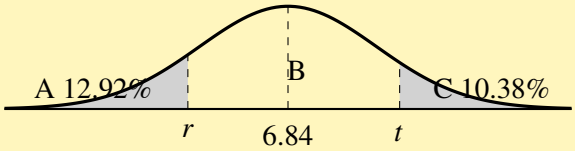
A1

TI-Nspire CX: Question 43

```
invNorm(0.8,0,1)
```

44 ← question *Worksheet · Q13*

(a)



A2(b) $r = \text{invNorm}(0.1292, 6.84, 0.25) = \boxed{6.56}$ **M1 A1**

$t = \text{invNorm}(0.8962, 6.84, 0.25) = \boxed{7.16}$ (left tail $1 - 0.1038$) **M1 A1**

A1

TI-Nspire CX: Question 44

```
invNorm(0.1292,6.84,0.25); invNorm(0.8962,6.84,0.25)
```

45 answer → Calculator **Worksheet · Q14**

[Maximum mark: 6]

The heights of certain flowers follow a normal distribution. It is known that 20% of these flowers have a height less than 3 cm and 10% have a height greater than 8 cm. Find the value of the mean μ and the standard deviation σ.

46 answer → Calculator **Worksheet · Q15**

[Maximum mark: 6]

The speeds of cars at a certain point on a straight road are normally distributed with mean μ and standard deviation σ. 15% of the cars travelled at speeds greater than 90 km h^{-1} and 12% of them at speeds less than 40 km h^{-1}. Find μ and σ.

45 ← question *Worksheet · Q14*

$z_1 = -0.8416$, $z_2 = 1.2816$ (left tail 0.9) **M1 A1 A1**

$3 = \mu - 0.8416\sigma,\ 8 = \mu + 1.2816\sigma$ **M1**

$\boxed{\sigma = 2.35}$, $\boxed{\mu = 4.98}$ **A1 A1**

TI-Nspire CX: Question 45

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

46 ← question *Worksheet · Q15*

$z_1 = \text{invNorm}(0.85) = 1.0364$, $z_2 = \text{invNorm}(0.12) = -1.1750$ **M1 A1 A1**

$90 = \mu + 1.0364\sigma,\ 40 = \mu - 1.1750\sigma$ **M1**

$\boxed{\sigma = 22.6}$, $\boxed{\mu = 66.6}$ **A1 A1**

TI-Nspire CX: Question 46

Two equations: menu → **3: Algebra** → **2: Solve System of Linear Equations**, e.g. $m + z_1 s = x_1$, $m + z_2 s = x_2$.

47 answer → Calculator **RV · Q10**

[Maximum mark: 5]

The heights of giraffes in a savanna can be modelled by a normal distribution with mean of 5.2 metres and standard deviation of $\sqrt{0.1}$ metres.

(a) Find the probability that a giraffe selected at random is at least 5 metres tall. [2]

(b) Twelve giraffes are independently selected at random and their heights recorded. Find the probability that more than nine of these giraffes are at least 5 metres tall. [3]

48 answer → Calculator **RV · Q23**

[Maximum mark: 6]

The weights W, in grams, of tea bags are normally distributed with a mean of 3.5 grams and a standard deviation of 0.53 grams. A tea bag is considered small if it weighs less than w grams.

(a) Given that 5.2% of tea bags are small, find w. [3]

(b) A selected tea bag is small. Find the probability that it weighs at least 2.25 grams. [3]

47 ← question *RV · Q10*

(a) $\mathrm{P}(H \geq 5) = \boxed{0.736}$ **M1 A1**

(b) $Y \sim \mathrm{B}(12, 0.736455)$: $\mathrm{P}(Y \geq 10) = \boxed{0.350}$ **M1 M1 A1**

TI-Nspire CX: Question 47

`normCdf(5,9E99,5.2,√(0.1)); binomCdf(12,0.736455,10,12)`

48 ← question *RV · Q23*

(a) $w = \text{invNorm}(0.052, 3.5, 0.53) = \boxed{2.64}$ g **M1 A2**

(b) $\dfrac{\mathrm{P}(2.25 \leq W < w)}{\mathrm{P}(W < w)} = \dfrac{0.052 - 0.009175}{0.052} = \boxed{0.824}$ **M1 M1 A1**

TI-Nspire CX: Question 48

`invNorm(0.052,3.5,0.53); normCdf(2.25,2.6384,3.5,0.53)/0.052`

49 answer → Calculator **RV · Q26**

[Maximum mark: 15]

A garden bed has a large number of flowers. The heights of the flowers are normally distributed with a mean of 43 cm and a standard deviation of 6 cm. A flower is classified as tall if it is taller than 48 cm.

(a) A flower is selected at random. (i) Find the probability that this flower is tall. (ii) Given that the flower is tall, find the probability it is taller than 55 cm. [6]

(b) Two flowers are selected at random. Find the probability they are both tall. [3]

(c) Five hundred flowers are selected at random. (i) Find the probability that at least 100 of these 500 flowers are tall. (ii) Find the expected number of tall flowers among these 500 flowers. [6]

49 ← question *RV · Q26*

(a)(i) $\boxed{0.202}$ (ii) $\dfrac{0.022750}{0.202328} = \boxed{0.112}$ **M1 A1 M1 M1 A2**

(b) $0.202328^2 = \boxed{0.0409}$ **M1 A2**

(c) $X \sim \mathrm{B}(500, 0.202328)$: (i) $\mathrm{P}(X \geq 100) = \boxed{0.569}$ (ii) $\mathrm{E}(X) = \boxed{101.2}$

(not rounded) **M1 A2 M1 A2**

TI-Nspire CX: Question 49

```
normCdf(48,9E99,43,6); binomCdf(500,0.202328,100,500)
```

50 answer → Calculator **RV · Q19**

[Maximum mark: 12]

The weights W, in grams, of frogs in an enclosure are normally distributed with a mean of 230 grams and a standard deviation of 15 grams. A frog that weighs less than 210 grams is considered as a small frog. A frog is selected at random.

(a) Find the probability that this frog is small. [3]

(b) A frog that weighs more than 250 grams is considered as a big frog. Another frog is selected at random. Find the probability that the frog is neither small nor big. [3]

(c) Frogs that aren't small or big are considered as standard frogs. A sample of 450 frogs is selected. Find the expected number of standard frogs in this sample. [3]

(d) Another sample of N frogs is selected. Find the minimum value of N such that the expected number of standard frogs in the sample is at least 100. [3]

50 ← question *RV · Q19*

(a) $\boxed{0.0912}$ **M1 A2**

(b) $\mathrm{P}(210 \leq W \leq 250) = \boxed{0.818}$ **M1 A2**

(c) $450 \times 0.817578 = \boxed{367.9}$ (do not round) **M1 A2**

(d) $0.817578N \geq 100 \Rightarrow N \geq 122.3$: $\boxed{N = 123}$ **M1 A2**

TI-Nspire CX: Question 50

```
normCdf(210,250,230,15)
```

51 answer → Calculator **RV · Q36**

[Maximum mark: 15]

The volumes, in mL, of olive oil bottles can be modelled by a normal distribution, with mean 508 mL and standard deviation 12 mL.

(a) Find the probability that a randomly selected bottle has a volume less than 500 mL. Round your answer to 3 significant figures. [2]

(b) The probability that a randomly selected bottle has a volume greater than v mL is 0.366. Find the value of v. [2]

(c) A bottle is randomly selected. Given that the bottle has a volume greater than 500 mL, find the probability that it has a volume greater than 520 mL. Round your answer to 3 significant figures. [3]

(d) From a random sample of 200 bottles, determine to the nearest integer the number of bottles that would be expected to have a volume within 2 standard deviations of the mean. [4]

(e) The bottles of olive oil are delivered to supermarkets in packs of 30. Determine the probability that at least 5 bottles from a randomly selected pack have a volume less than 495 mL. Round your answer to 3 significant figures. [4]

51 ← question *RV · Q36*

(a) $\boxed{0.252}$ **M1 A1**

(b) left tail 0.634: $v = \boxed{512}$ **M1 A1**

(c) $\dfrac{0.158655}{0.747508} = \boxed{0.212}$ **M1 M1 A1**

(d) $\mathrm{P}(484 < X < 532) = 0.9545$; $200 \times 0.9545 = 190.9 \approx \boxed{191}$ (question asks for an integer) **M1 A1 M1 A1**

(e) $p = 0.13933$; $\mathrm{B}(30, p)$: $\mathrm{P}(Y \geq 5) = \boxed{0.409}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 51

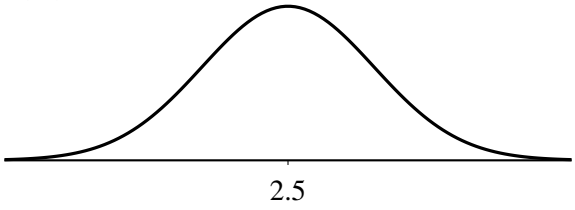
```
invNorm(0.634,508,12); binomCdf(30,0.13933,5,30)
```

52 answer → Calculator **Worksheet · Q16**

[Maximum mark: 13]

The weights of chickens for sale in a shop are normally distributed with mean 2.5 kg and standard deviation 0.3 kg.

(a) A chicken is chosen at random. (i) Find the probability that it weighs less than 2 kg. (ii) Find the probability that it weighs more than 2.8 kg. (iii) Copy the diagram below. Shade the areas that represent the probabilities from parts (i) and (ii).



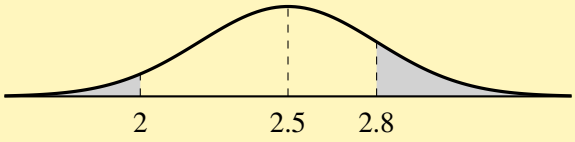
(iv) Hence show that the probability that it weighs between 2 kg and 2.8 kg is 0.7936 (to four significant figures). [7]

(b) A customer buys 10 chickens. (i) Find the probability that all 10 chickens weigh between 2 kg and 2.8 kg. (ii) Find the probability that at least 7 of the chickens weigh between 2 kg and 2.8 kg. [6]

52 ← question *Worksheet · Q16*

(a)(i) $\boxed{0.0478}$ (ii) $\boxed{0.159}$ **A2 A2**

(iii)



A1(iv) $1 - 0.04779 - 0.15866 = 0.79355 = 0.7936$ **M1 A1**

(b) $Y \sim \mathrm{B}(10, 0.79355)$: (i) $0.79355^{10} = \boxed{0.0990}$ (ii) $\mathrm{P}(Y \geq 7) = \boxed{0.867}$

M1 A2 M1 A2

TI-Nspire CX: Question 52

```
normCdf(2,2.8,2.5,0.3); binomCdf(10,0.79355,7,10)
```

53 answer → Calculator **Worksheet · Q17**

[Maximum mark: 13]

A van can take either Route A or Route B for a particular journey. If Route A is taken, the journey time may be assumed to be normally distributed with mean 46 minutes and a standard deviation 10 minutes. If Route B is taken, the journey time may be assumed to be normally distributed with mean μ minutes and standard deviation 12 minutes.

(a) For Route A, find the probability that the journey takes more than 60 minutes. [2]

(b) For Route B, the probability that the journey takes less than 60 minutes is 0.85. Find the value of μ. [3]

(c) The van sets out at 06:00 and needs to arrive before 07:00. (i) Which route should it take? (ii) Justify your answer. [3]

(d) On five consecutive days the van sets out at 06:00 and takes Route B. Find the probability that (i) it arrives before 07:00 on all five days; (ii) it arrives before 07:00 on at least three days. [5]

53 ← question *Worksheet · Q17*

(a) $\boxed{0.0808}$ **M1 A1**

(b) $z = 1.0364$; $\mu = 60 - 12(1.0364) = \boxed{47.6}$ **M1 A1 A1**

(c) **Route A**: $\mathrm{P}(A < 60) = 0.919 > 0.85 = \mathrm{P}(B < 60)$ **A1 R2**

(d) $Y \sim \mathrm{B}(5, 0.85)$: (i) $0.85^5 = \boxed{0.444}$ (ii) $\mathrm{P}(Y \geq 3) = \boxed{0.973}$ **M1 A1**

M1 A2

TI-Nspire CX: Question 53

```
normCdf(60,9E99,46,10); invNorm(0.85,0,1); binomCdf(5,0.85,3,5)
```

54 answer → Calculator **Worksheet · Q18**

[Maximum mark: 17]

In a large school, the heights of all fourteen-year-old students are measured. The heights of the girls are normally distributed with mean 155 cm and standard deviation 10 cm. The heights of the boys are normally distributed with mean 160 cm and standard deviation 12 cm.

(a) Find the probability that a girl is taller than 170 cm. [3]

(b) Given that 10% of the girls are shorter than x cm, find x. [3]

(c) Given that 90% of the boys have heights between q cm and r cm where q and r are symmetrical about 160 cm, and $q < r$, find the value of q and of r. [4]

In the group of fourteen-year-old students, 60% are girls and 40% are boys. The probability that a girl is taller than 170 cm was found in part (a). The probability that a boy is taller than 170 cm is 0.202. A fourteen-year-old student is selected at random.

(d) Calculate the probability that the student is taller than 170 cm. [4]

(e) Given that the student is taller than 170 cm, what is the probability the student is a girl? [3]

54 ← question *Worksheet · Q18*

(a) $\boxed{0.0668}$ **M1 A2**

(b) $x = \text{invNorm}(0.1, 155, 10) = \boxed{142}$ **M1 A2**

(c) 5% in each tail: $q = \boxed{140}$, $r = \boxed{180}$ **M1 A1 A1 A1**

(d) $0.6(0.0668) + 0.4(0.202) = \boxed{0.121}$ **M1 A1 M1 A1**

(e) $\dfrac{0.6 \times 0.0668}{0.12088} = \boxed{0.332}$ **M1 A1 A1**

TI-Nspire CX: Question 54

```
normCdf(170,9E99,155,10); invNorm(0.05,160,12),invNorm(0.95,160,12)
```

55 answer → Calculator **Worksheet · Q19**

[Maximum mark: 15]

The weights of players in a sports league are normally distributed with a mean of 76.6 kg (correct to three significant figures). It is known that 80% of the players have weights between 68 kg and 82 kg. The probability that a player weighs less than 68 kg is 0.05.

(a) Find the probability that a player weighs more than 82 kg. [2]

(b) (i) Write down the standardized value, z, for 68 kg. (ii) Hence, find the standard deviation of weights. [4]

To take part in a tournament, a player's weight must be within 1.5 standard deviations of the mean.

(c) (i) Find the set of all possible weights of players that take part in the tournament. (ii) A player is selected at random. Find the probability that the player takes part in the tournament. [5]

Of the players in the league, 25% are women. Of the women, 70% take part in the tournament.

(d) Given that a player selected at random takes part in the tournament, find the probability that the selected player is a woman. [4]

55 ← question *Worksheet · Q19*

(a) $1 - 0.8 - 0.05 = \boxed{0.15}$ **M1 A1**

(b)(i) $z = \boxed{-1.64}$ (-1.645) (ii) $\sigma = \dfrac{8.6}{1.645} = \boxed{5.23}\,\text{kg}$ **A1 M1 A2**

(c)(i) $76.6 \pm 1.5(5.228)$: $\boxed{68.8 < W < 84.4}$ **M1 A1 A1** (ii) $\boxed{0.866}$ **M1 A1**

(d) $\dfrac{0.25 \times 0.7}{0.8664} = \boxed{0.202}$ **M1 A1 M1 A1**

TI-Nspire CX: Question 55

```
invNorm(0.05,0,1); normCdf(68.76,84.44,76.6,5.228)
```

56 answer → Calculator **Oxford Exam-style · Q21**

[Maximum mark: 7]

A packing machine produces bags of rice whose weights are normally distributed with mean 50.1 kg and standard deviation 0.4 kg. If a bag produced by this machine is selected at random, find the probability that its weight is:

(a) less than 49.5 kg [2]

(b) between 49.5 kg and 50.5 kg [2]

(c) more than 49 kg given that it is less than 49.5 kg. [3]

57 answer → Calculator **Oxford Exam-style · Q25**

[Maximum mark: 12]

A national park houses a large number of tall trees. The heights of the trees are normally distributed with mean of 45 metres and a standard deviation of 9 metres.

(a) One tree is selected at random. (i) Find the probability that this tree is at least 55 metres high. (ii) Given that this tree is at least 55 metres, find the probability that it is taller than 65 metres. [5]

(b) Three trees are selected at random. Find the probability that they are all at least 55 metres high. [2]

(c) A selection of 50 trees is made at random. (i) Find the expected number of these trees that are taller than 55 metres. (ii) Find the probability that at least 5 of these trees are taller than 55 metres. [5]

56 ← question *Oxford Exam-style · Q21*

(a) $\boxed{0.0668}$ **M1 A1**

(b) $\boxed{0.775}$ **M1 A1**

(c) $\dfrac{\mathrm{P}(49 < X < 49.5)}{\mathrm{P}(X < 49.5)} = \boxed{0.955}$ **M1 M1 A1**

TI-Nspire CX: Question 56

```
normCdf(49,49.5,50.1,0.4)/normCdf(-9E99,49.5,50.1,0.4)
```

57 ← question *Oxford Exam-style · Q25*

(a)(i) $\boxed{0.133}$ (ii) $\dfrac{0.013134}{0.133260} = \boxed{0.0986}$ **M1 A1 M1 M1 A1**

(b) $0.133260^3 = \boxed{0.00237}$ **M1 A1**

(c) $N \sim \mathrm{B}(50, 0.133260)$: (i) $\boxed{6.66}$ (ii) $\mathrm{P}(N \geq 5) = \boxed{0.814}$ **M1 A1 M1 M1 A1**

TI-Nspire CX: Question 57

```
normCdf(55,9E99,45,9); binomCdf(50,0.13326,5,50)
```

58 answer → Calculator SME Hard · Q10

[Maximum mark: 8]

(a) The running time of films is normally distributed with a mean time of 102 minutes and a standard deviation of 13 minutes. Find the probability that, on a randomly selected day, the feature film playing at the cinema has a running time of between 97 and 108 minutes. [2]

(b) Jonah watches a film on 18 different occasions. Find the expected number of occasions on which the film he watches will last less than 95 minutes. [3]

(c) Find the probability that on at least 6 out of the 18 occasions, the film will last for longer than 99 minutes. [3]

59 answer → Calculator SME Very Hard · Q8

[Maximum mark: 10]

(a) Alannah cycles to school each day via one of two possible routes and the time taken to complete either journey follows a normal distribution. The journey time for route A has a mean of 28 minutes and a standard deviation of 10 minutes. The journey time for route B has a mean of 33 minutes and a standard deviation of 4 minutes. Identify an advantage of each route. [2]

(b) The school day begins at 08:30 and Alannah leaves her house at 07:55. Determine which route is more likely to make her late. [3]

(c) Route A is closed off for maintenance work so Alannah travels by route B every day from Monday to Friday. Find the probability that she is on time for school on 2 consecutive days and late for the other 3 days. [5]

58 ← question *SME Hard · Q10*

(a) $\boxed{0.328}$ **M1 A1**

(b) $18 \times 0.295129 = \boxed{5.31}$ **M1 M1 A1**

(c) $p = \mathrm{P}(T > 99) = 0.591$; $\mathrm{B}(18, p)$: $\mathrm{P}(X \geq 6) = \boxed{0.993}$ **M1 M1 A1**

TI-Nspire CX: Question 58

```
normCdf(97,108,102,13); binomCdf(18,0.59125,6,18)
```

59 ← question *SME Very Hard · Q8*

(a) A: shorter mean time; B: smaller s.d. — more reliable **A1 A1**

(b) late if $T > 35$: A 0.242, B 0.309: **Route B** **M1 A1 A1**

(c) on time $p = 0.691462$; 4 consecutive positions (MT, TW, WTh, ThF): $4p^2(1-p)^3 = \boxed{0.0562}$ **M1 M1 M1 A2**

TI-Nspire CX: Question 59

```
normCdf(35,9E99,33,4)
```

60 answer → Calculator **SME Very Hard · Q10**

[Maximum mark: 14]

(a) The time spent on a smart phone per day by teenage boys is normally distributed with a mean of 306 minutes and a variance of 56^2. Find the proportion of teenage boys that spend between 275 minutes and 375 minutes per day on a smart phone. Sketch a diagram to show this information. [4]

(b) A sample of 50 teenage boys are selected from this population. Find the expected number of boys from the sample who spend more than 7 hours on their smart phone each day. [3]

(c) The time spent on smart phones each day by teenage girls is normally distributed with a mean of 286 minutes and a standard deviation of 74 minutes. 25 teenage girls are selected at random. Find the probability that exactly 3 of them spend less than 200 minutes on a smart phone each day. [3]

(d) A teenager is selected at random from the group of 50 boys in part (b) and the group of 25 girls in part (c). Find the probability that the teenager is a boy, given that they spend less than 250 minutes on a smart phone each day. [4]

60 ← question *SME Very Hard · Q10*

(a) $\boxed{0.601}$ (sketch: curve at 306, shade 275 to 375) **A2 M1 A1**

(b) 7 h = 420 min: $50 \times 0.020889 = \boxed{1.04}$ **M1 M1 A1**

(c) $p = 0.12258$; $\mathrm{P}(X = 3) = \boxed{0.239}$ **M1 M1 A1**

(d) $\dfrac{50(0.15866)}{50(0.15866)+25(0.31331)} = \boxed{0.503}$ **M1 M1 M1 A1**

TI-Nspire CX: Question 60

```
normCdf(275,375,306,56); binomPdf(25,0.12258,3)
```